

ATPL EASA STUDY GUIDE | 040 - HUMAN PERFORMANCE

Chapter 11: Stress & Workload Management

Syllabus Source: CAE Oxford (pp. 355-392) | Pass Target: >= 90% AviationExam

1. AROUSAL & THE YERKES-DODSON LAW (INVERTED-U CURVE)

Human performance varies non-linearly with the level of psychological and physiological arousal (the Yerkes-Dodson Law):

Arousal State	Cockpit Environmental Conditions	Performance & Cognitive Characteristics
Hypo-Arousal (Underload / Boredom)	Low stress, low stimulation (long cruise at night).	Inattention, sluggish reaction time, sleepiness, complacency, high error rate.
Optimum Arousal (Peak Performance)	Moderate stress, challenging but manageable workload.	MAXIMUM cognitive performance, sharp sensory perception, flexible reasoning, optimal situational awareness.
Hyper-Arousal (Overload / Panic)	Extreme acute stress, severe emergency, sensory overload.	Catastrophic breakdown in performance, cognitive tunneling, motor regression, paralysis of action ('freezing').

2. HANS SELYE'S GENERAL ADAPTATION SYNDROME (GAS)

The human biological response to prolonged stressors proceeds through three predictable physiological stages:

- 1. Alarm Reaction: Acute 'fight-or-flight' activation via the sympathetic nervous system and adrenal glands. Release of adrenaline and noradrenaline. Increased heart rate, blood pressure, pupil dilation, muscle tension.
- 2. Resistance Stage: The body adapts to sustained stress. Cortisol release maintains high blood glucose levels. Parasympathetic system moderates shock. High metabolic energy expenditure.
- 3. Exhaustion Stage: Prolonged chronic stress drains bodily reserves. Immune breakdown, burnout, clinical depression, chronic fatigue, severe cognitive degradation.

3. ACUTE VS CHRONIC STRESS

- Acute Stress: Short-term, sudden demand (e.g. engine failure at V1, windshear encounter). Mobilizes body resources immediately. Resolves once crisis ends.
- Chronic Stress: Long-term persistent cumulative pressure (marital conflict, bereavement, financial troubles, toxic workplace). Insidious: permanently raises baseline cortisol, degrades sleep quality, and lowers tolerance to acute in-flight crises.

4. COCKPIT WORKLOAD MANAGEMENT & TASK PRIORITIZATION

[!] EXAM TRAP (AVIATIONEXAM) - The Universal Cockpit Golden Rule: Aviate - Navigate - Communicate

When dealing with any emergency, high workload, or system malfunction, the flight crew must strictly enforce the golden hierarchy: 1. AVIATE: Maintain control of the aircraft, pitch attitude, airspeed, and altitude; 2. NAVIGATE: Know present position, terrain clearance (MSA), and heading; 3. COMMUNICATE: Advise ATC, company, and cabin crew only AFTER the aircraft is stabilized; 4. ADMINISTER: Complete systems diagnostics, ECAM/QRH checklists, and passenger briefings.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Algoritmo Paso a Paso (Guía para Dummies): SHELL Model & Error Analysis

REGLA MNEMOTÉCNICA: L-H-S-E-L Paso 1: Identificar siempre el Liveware Central (L) como el piloto/operador humano. Paso 2: Si el problema involucra mandos, pantallas o botones confusos -> Es un desajuste L-H (Hardware). Paso 3: Si involucra una checklist mal redactada o un manual confuso -> Es un desajuste L-S (Software). Paso 4: Si involucra turbulencia, calor o hipoxia en cabina -> Es un desajuste L-E (Environment). Paso 5: Si involucra mala coordinación entre Capitán y Primer Oficial o con ATC -> Es un desajuste L-L (Liveware).

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What does the Yerkes-Dodson Law state regarding the relationship between arousal and human performance?

- [A]: Performance increases linearly with stress indefinitely
- [B]: Performance is an inverted U-curve: optimal at intermediate arousal, but poor under both under-arousal and over-arousal
- [C]: Performance is independent of psychological stress
- [D]: Pilots perform best under extreme panic

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. The inverted-U curve demonstrates that both sensory underload (boredom) and severe overload (panic) degrade performance; peak capability occurs at moderate arousal.

Q2: In Hans Selye's General Adaptation Syndrome (GAS), which stage immediately follows the initial Alarm Reaction?

- [A]: The Exhaustion stage
- [B]: The Resistance stage
- [C]: The Recovery stage
- [D]: The Burnout stage

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. The three phases of GAS are: 1) Alarm Reaction, 2) Resistance, and 3) Exhaustion.

Q3: When multiple alarms and flight anomalies occur simultaneously, what is the mandatory priority order of actions?

- [A]: Communicate -> Navigate -> Aviate
- [B]: Aviate -> Navigate -> Communicate
- [C]: Checklists -> Radios -> Flying
- [D]: Declare MAYDAY -> Read QRH -> Hand fly

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. 'Aviate, Navigate, Communicate' is the absolute priority rule in aviation safety.

Q4: How does severe acute stress affect a pilot's visual and mental processing?

- [A]: Expands peripheral field of view

- [B]: Induces cognitive tunneling (attentional narrowing onto a single feature, ignoring critical warnings)
- [C]: Enhances ability to perform complex mathematics
- [D]: Eliminates spatial disorientation

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Severe stress causes perceptual and cognitive tunneling, blinding the pilot to peripheral cues and multi-task information.

Q5: What is the primary operational hazard of cockpit underload (hypo-arousal) during long oceanic flights?

- [A]: Adrenaline surge
- [B]: Complacency, vigilance decrement, loss of situational awareness, and delayed response to unexpected events
- [C]: Immediate hyperventilation
- [D]: Otic barotrauma

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Low workload leads to complacency and reduced vigilance, leaving crew unprepared for sudden emergency deviations.

Q6: Which biological system triggers the immediate 'fight-or-flight' alarm response during an engine fire on takeoff?

- [A]: The parasympathetic nervous system
- [B]: The sympathetic nervous system and adrenal medulla releasing adrenaline
- [C]: The vestibular otolith organs
- [D]: The immune system

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. The sympathetic nervous system triggers instant adrenaline release, elevating heart rate and blood pressure for rapid physical response.